

DEVA ANAND

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Education

MS, Computer Science, University of Massachusetts Amherst 2025–2027

Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

BE, Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies 2019–2023

Professional Experience

TherAlign Health, Founding Software Engineer Intern | *Remote* Jul 2026 – Present

- Founding engineer building full-stack features (Next.js/React front end, Firebase/GCP backend) for a provider-facing **SMART on FHIR** clinical decision support app that integrates AI into a real-world prescribing workflow.
- Extending a **Firebase Cloud Functions** backend that builds medication alternatives from **RxNorm** normalization, formulary and prior-authorization logic, and PubMed/guideline evidence, with **Gemini** constrained to synthesis while the backend deterministically controls eligibility, keeping the generative output grounded and verifiable.

Culture & Morality Lab, UMass Amherst, Lab Assistant | *Amherst, MA* Jul 2026 – Present

- Sole engineer building the lab's text-analysis platform (**Python**, **FastAPI**, sentence-transformers, React), productionizing a published NLP method into a self-serve web tool with full ownership of architecture and technical decisions delegated by the PI.
- Built a YAML model registry, a versioned construct library (**99 constructs from 38 questionnaires**), and structured data-quality warnings; grew the backend suite to **40 tests** running hermetically in CI.

Wysa, Backend Engineer | *Bengaluru, India* Jul 2023 – Jul 2025

- Engineered a multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clinical clients**, processing **10,000+ monthly triage submissions** via Mayden iaptus **REST** integrations; contributed to **\$340K+** in revenue.
- Architected a full-stack annotation platform (**React**, Node.js, MongoDB) used daily by the AI team via secure **REST APIs**, cutting manual annotation by **100+ hours/month** and lifting labeling throughput by **60%**.
- Diagnosed and resolved production issues across tenants, monitoring logs during releases and client integrations and shipping targeted fixes for tenant-specific incidents to keep the multi-tenant eTriage service stable.

Projects

AgenticSearch – Provenance-First Entity Discovery (GitHub | Demo) Python, FastAPI, OpenAI / Groq, SQLite

- Built a multi-stage agentic web-discovery service (Python, FastAPI) that turns open-ended queries into structured entity tables via typed retrieval planning, reranking, and deterministic extraction with **LLM** fallback and cell-level provenance; hardened with **150+ automated tests** and deployed on **Google Cloud Run**.

cf-ai-research-scout – Edge-Native AI Assistant (GitHub) TypeScript, Cloudflare Workers AI, Durable Objects, D1

- Built an edge-native **AI** research assistant end to end in **TypeScript** on Cloudflare Workers, where an agent fetches web sources, generates multi-step research digests, and streams responses to the client over WebSocket, with durable state persisted in D1.

AutoEval – Retrieval Evaluation Improvement Agent (GitHub) Python, IR evaluation, Git automation

- Built a Python agent that automates evaluation-suite improvement, proposing queries and labels, scoring against multiple systems, and committing only accepted updates back to **Git** with report cards, guarded by schema checks and read-only holdout protection.

Publications

- "Alzheimer's Disease Classification using Transfer Learning," IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

Skills

Languages: Python, JavaScript, TypeScript, HTML/CSS, SQL, Java, Bash

AI & LLM: LLM agents, RAG, prompt engineering, model evaluation, OpenAI, Gemini, Groq

Frameworks & Web: FastAPI, Node.js, Express, React, Next.js, REST APIs

ML & Data: PyTorch, Hugging Face Transformers, sentence-transformers, pandas, NumPy

Cloud & DevOps: Azure, AWS, GCP, Docker, Git, GitHub Actions, CI/CD

Open Source & Community

- Selected for **GirlScript Summer of Code 2023**; contributed to Linkfree, Freehit, and ProjectsHut. - Tutored **30+ underprivileged students** in programming at Sayur, a non-profit in Tamil Nadu.